VLSM (Variable Length Subnet Mask) Address Plan for 50 Rural Health Clinics

HealthReach Foundation | Batch 17

Submission detail	Information
Institution	MIVA Open University
Course	MIT 8105 Network Architecture and Protocols
Student	AJIBOLA AFOLABI
Student email	a.afolabi8643@miva.edu.ng

Executive overview

This report sets out an IPv4 address plan for 50 HealthReach clinics across Rivers, Benue and Kaduna. All clinic addresses come from one pool, 10.80.0.0/21, divided by region: Rivers takes 10.80.0.0/22, Benue 10.80.4.0/23 and Kaduna 10.80.6.0/23. Inside those blocks, the 10 large clinics get /26 networks, the 20 medium clinics /27 and the 20 small clinics /28. Because each region is a single aligned block, the cloud needs only three summary routes instead of 50 clinic prefixes, and a clinic keeps the same addresses on cellular or satellite.

1 Scenario interpretation and planning assumptions

HealthReach provides vaccination and maternal-health services in three Nigerian states. Clinic tablets and cold-chain sensors synchronise with a regional hub and cloud dashboard over 3G/4G or satellite links. The scenario gives no counts, so these are my own assumptions, to be confirmed by site survey: Rivers 20 clinics (RV01–RV20), Benue 15 (BN01–BN15) and Kaduna 15 (KD01–KD15); 10 large, 20 medium and 20 small clinics needing 50, 24 and 10 addressed interfaces, including gateway and growth allowance. The plan covers IPv4 clinic LANs only, one subnet per clinic, with hubs in a separate 10.81.0.0/24 pool.

2 Addressing principles and VLSM calculations

VLSM lets one plan use subnets of different sizes. For each clinic I chose the smallest ordinary LAN subnet that fits its host requirement and placed the larger subnets first so every block stays aligned. CIDR aggregation then lets aligned allocations be advertised as one shorter prefix (Fuller & Li, 2006). The pool sits inside the private 10.0.0.0/8 range (Rekhter et al., 1996), so cloud and partner networks must be checked for overlap.

2.1 Host sizing rule

Usable hosts equal $2^{(32 - \text{prefix length})} - 2$. For a demand H, I find the smallest h where $2^h - 2 \geq H$; the prefix is 32 − h.

Clinic class	Demand	Calculation	Prefix and mask	Usable	Unused usable
Large	50	$2^6 - 2 = 62$	/26, 255.255.255.192	62	12
Medium	24	$2^5 - 2 = 30$	/27, 255.255.255.224	30	6
Small	10	$2^4 - 2 = 14$	/28, 255.255.255.240	14	4

A /27 gives only 30 usable addresses, too few for 50, and a /29 gives six, too few for 10. A /31 suits point-to-point links only (Retana et al., 2000). Networks advance in steps of 64 (/26), 32 (/27) and 16 (/28), so none overlap.

3 Regional hierarchy and route summaries

Region	Clinic mix (L/M/S)	Clinic addresses	Regional summary	Full covered range
Rivers	4 / 8 / 8	640	10.80.0.0/22	10.80.0.0–10.80.3.255
Benue	3 / 6 / 6	480	10.80.4.0/23	10.80.4.0–10.80.5.255
Kaduna	3 / 6 / 6	480	10.80.6.0/23	10.80.6.0–10.80.7.255
Total	10 / 20 / 20	1,600	10.80.0.0/21	10.80.0.0–10.80.7.255

Rivers needs 4 × 64 + 8 × 32 + 8 × 16 = 640 addresses, so the smallest covering block is a /22 (1,024). Benue and Kaduna each need 480, so a /23 (512) fits. Together the blocks fill the /21 exactly (2,048).

3.1 Why the summaries are valid

The blocks are aligned, do not overlap and sit inside 10.80.0.0/21. In binary, third octets 0 to 3 share six bits, 4 to 5 and 6 to 7 share seven, and 0 to 7 share five. The summaries record who owns a range, not that every address is in use, so they stay stable when a single clinic link fails.

3.2 Spare regional blocks

The 448 unallocated addresses are 384 in Rivers (10.80.2.128/25 and 10.80.3.0/24) and 32 each in Benue (10.80.5.224/27) and Kaduna (10.80.7.224/27). Benue and Kaduna cannot add a new /26 without extra space. If surveys show faster expansion, extra blocks should be reserved outside this /21 rather than stretching a summary into another region's space.

4 Clinic subnet schedule

The default gateway is the first usable address of each subnet and the last is reserved for the local sync server. Each row covers consecutive clinics, so all 50 subnets are listed.

Clinics	Prefix	Network addresses (step)
RV01–RV04	/26	10.80.0.0 to 10.80.0.192 (64)
RV05–RV12	/27	10.80.1.0 to 10.80.1.224 (32)
RV13–RV20	/28	10.80.2.0 to 10.80.2.112 (16)
BN01–BN03	/26	10.80.4.0 to 10.80.4.128 (64)
BN04–BN09	/27	10.80.4.192 to 10.80.5.96 (32)
BN10–BN15	/28	10.80.5.128 to 10.80.5.208 (16)
KD01–KD03	/26	10.80.6.0 to 10.80.6.128 (64)
KD04–KD09	/27	10.80.6.192 to 10.80.7.96 (32)
KD10–KD15	/28	10.80.7.128 to 10.80.7.208 (16)

As a worked example, BN10 is on 10.80.5.128/28. Its gateway is 10.80.5.129, its sync server can use 10.80.5.142 and its broadcast address is 10.80.5.143. Its route belongs only to the Benue summary, 10.80.4.0/23.

5 Hub addressing and regional routing

Hubs and cloud services use the separate pool 10.81.0.0/24: River's hub 10.81.0.0/27, Benue hub 10.81.0.32/27, Kaduna hub 10.81.0.64/27 and cloud services 10.81.0.128/26. Each regional hub keeps individual clinic routes, while the cloud VPN router holds only three.

Cloud route	Logical next hop	Scope
10.80.0.0/22	Rivers regional VPN peer	RV01–RV20 and reserve
10.80.4.0/23	Benue regional VPN peer	BN01–BN15 and reserve
10.80.6.0/23	Kaduna regional VPN peer	KD01–KD15 and reserve

5.1 Aggregate safety and failure handling

Static routes with health tracking are enough at this scale. Each regional router should carry a discard route for its summary, so longest-prefix matching prefers an active /26, /27 or /28 and traffic to reserved or withdrawn space is dropped instead of looping to the cloud (Fuller & Li, 2006). If a hub or its tunnel fails, the cloud route should be withdrawn using health checks. Surviving a hub failure needs a redundant hub, because clinic satellite backup does not remove that dependency.

5.2 Cellular primary and satellite backup

The clinic LAN prefix is identical on both transports. The clinic starts the authenticated VPN itself, reachability is tracked beyond the local modem, traffic moves to satellite only after sustained cellular failure, and a recovery delay prevents flapping.

6 Privacy, security and NDPR compliance

6.1 Privacy controls around the address plan

Neither RFC 1918 addressing nor VLSM encrypts patient records. Clinic-to-hub traffic should use authenticated IPsec VPNs (Kent & Seo, 2005), with application-level encryption for synchronisation. Firewalls deny by default: sync clients reach only their designated hub and local services, clinic-to-clinic traffic is blocked, and management access comes only from authorised administrator networks. Each sensor has its own credentials. One subnet per clinic does not separate sensors from tablets, so stronger separation adds clinical, IoT and management VLAN subnets after revising the device budget.

6.2 NDPR compliance and risk mitigation

Vaccination and maternal-health records are sensitive personal data. The Nigeria Data Protection Regulation (NDPR) and the Nigeria Data Protection Act 2023 expect security safeguards, data minimisation and breach response, and ISO/IEC 27001 alignment supports this.

Risk	Mitigation
Interception on cellular or satellite links	Authenticated IPsec VPNs plus application-level encryption
Lateral movement between clinics	Default-deny firewalls and blocked clinic-to-clinic traffic
Device theft or unauthorised administration	Per-device credentials, encrypted tablets, restricted management networks

Governance rests on an address register recording each clinic's prefix, gateway, reservations, owner and approvals. Offline patient data must stay available to authorised staff only.

6.3 Address administration and offline resilience

DHCP runs on each clinic router, with recorded reservations excluded from the dynamic pool, so tablets keep working when the WAN fails. Local DNS points to the sync server and no cloud addresses are hard-coded. Records are stored encrypted and queued offline, then synchronised with acknowledgements, retries and duplicate detection. Cold-chain gateways raise temperature alarms locally.

7 Efficiency assessment

Metric	Result	Interpretation
Allocated clinic subnet addresses	1,600	10 × 64 + 20 × 32 + 20 × 16
Usable clinic host addresses	1,500	1,600 less 100 network and broadcast
Design demand including gateways	1,180	10 × 50 + 20 × 24 + 20 × 10
Host capacity utilisation	78.67%	1,180 ÷ 1,500 × 100
Unallocated regional reserve	448	2,048 − 1,600

Metric	Result	Interpretation
Upstream clinic route reduction	94%	(50 − 3) ÷ 50 × 100

A /26 for every clinic would use 3,200 addresses against 1,600 for VLSM, a saving of 50%. A /24 per clinic would use 12,800, a saving of 87.5%. Utilisation is planned, not measured on live devices.

8 Validation

The plan was checked as follows: 20 + 15 + 15 = 50 distinct clinics; all networks are disjoint and correctly aligned; each gateway is a valid first usable address; all summaries fit 10.80.0.0/21; and 640 + 480 + 480 + 448 = 2,048. Testing should confirm satellite failover without renumbering and local operation with both links down.

9 Conclusion

The plan delivers the required 50-clinic VLSM schedule using a 10.80.0.0/21 pool, three regional aggregates and /26, /27 and /28 networks sized to the assumed demand. Its main limitations are the assumed inventory, the limited growth space in Benue and Kaduna, and the single-subnet baseline at each clinic. These should be confirmed before operation.

References

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